Mentorship App#1: Algorithmic Hype-Check Financial & Technology Research Dashboard

Mentor: Michael Mannarino
Mentee: Isabella
Cadence: 2–4 hours/week of direct mentorship + independent project execution [cite: 2].

Project Overview & Objectives

Problem Statement

Retail investors and technology enthusiasts are exposed to an overwhelming volume of financial news, social media discussions, and market narratives. While information is more accessible than ever, users often spend significant time researching the same story across multiple sources, determining which claims are evidence-based, and identifying what information is actually relevant. This environment also makes it easier for speculative narratives and hype-driven discussions to spread faster than supporting evidence. The result is increased research burden, fragmented context, and difficulty distinguishing between compelling stories and well-supported theses.

Core Product Vision

We are building an AI-powered research dashboard that helps users understand important financial and technology developments without requiring them to read dozens of articles, forum discussions, and reports. Rather than providing investment recommendations or price predictions, the platform aggregates information, identifies major claims being made, evaluates the evidence supporting those claims, and presents concise summaries with optional deeper analysis. The goal is to reduce research effort while preserving the user's ability to make independent decisions.

MVP Scope

- Users can search for a public company, stock ticker, cryptocurrency, or technology topic.
- The platform will automatically collect relevant articles and discussions from selected news and community sources.
- AI models will:
    - Generate concise summaries of major developments.
    - Extract key claims and narratives from articles.
    - Identify supporting evidence mentioned within the content.
    - Highlight missing information or unresolved questions.
    - Calculate a Hype vs. Evidence score based on factors such as technical specificity, supporting data, benchmarks, financial disclosures, and speculative language.

- Users will be able to view: Recent news and discussions, AI-generated summaries, Hype vs. Evidence metrics, Key claims and supporting evidence, and Historical changes in narrative and sentiment over time.

Out of Scope (Initial Version)

Stock recommendations, buy/sell signals, price predictions, portfolio management, automated trading, and personalized financial advice.

Technical Architecture Execution Matrix

Component	Technical Execution Details
Data Ingestion Pipeline	Scheduled cloud jobs collect articles and discussion data from approved news feeds and public APIs. Content is parsed, cleaned, and stored in a relational database.
AI Analysis Layer	LLMs perform summarization, claim extraction, evidence detection, sentiment analysis, entity recognition, and hype classification. Results are stored for retrieval and visualization.
Backend Services	API layer handles search, data retrieval, AI analysis requests, and historical trend aggregation.
Frontend Dashboard	Users search companies or topics and view summaries, evidence breakdowns, narrative trends, and hype metrics through a web interface.

Structured 12-Week Learning Plan

This customized roadmap maps the *Algorithmic Hype-Check Dashboard* features to Isabella's 12-week development schedule [cite: 1]. Special **Deep-Dive Learning Modules** have been injected to address the targeted high-priority conceptual milestones [cite: 44].

Phase 1: Foundations of the Cloud & Modern Data (Weeks 1–4) [cite: 4]

Goal: Shift from writing local scripts to building decoupled cloud systems [cite: 5].

- **Week 1: The Paradigm Shift (Local to Cloud) [cite: 6]**
    - *Concept:* Virtualization, compute vs. storage, and the shared responsibility model

[cite: 7].
    - *Hands-on Project Application:* Set up the foundational AWS/GCP account for the Hype-Check platform. Configure the local developer CLI environment and securely launch an ephemeral cloud instance to experiment with raw ingestion testing [cite: 8].
- **Week 2: Data Modeling & Storage Strategy [cite: 9]**
    - *Concept:* Structured vs. Unstructured data [cite: 10]. Choosing between Relational (PostgreSQL) for structured metric history vs. NoSQL (MongoDB/DynamoDB) vs. Object Storage (S3) for raw article text [cite: 10].
    - *Hands-on Project Application:* Design a relational database schema optimized to track tickers, extracted claims, historical sentiment trends, and source metadata [cite: 11]. Spin up a managed cloud database instance (e.g., RDS PostgreSQL) [cite: 11].
- **Week 3: Data Ingestion & API Integration [cite: 12]**
    - *Concept:* REST APIs, JSON data payloads, and event-driven architectures [cite: 13].
    - *Hands-on Project Application:* Write a clean, containerized Python script using Docker [cite: 14]. This script will fetch live news feeds and discussion board API text data, transforming the JSON payload to stream directly into raw cloud storage [cite: 14].
- **Week 4: Serverless Architectures [cite: 15]**
    - *Concept:* Microservices and event triggers, shifting away from always-on hosting [cite: 16].
    - *Hands-on Project Application:* Refactor the Week 3 news ingestion script into an on-demand serverless function (AWS Lambda or GCP Cloud Function) triggered by a cron-timer to keep the dashboard refreshed automatically [cite: 17].

Phase 2: Demystifying AI & LLM Orchestration (Weeks 5–8) [cite: 18]

Goal: Learn how to leverage state-of-the-art AI programmatically without training models from scratch [cite: 19].

- **Week 5: The AI Landscape (APIs over Training) [cite: 20]**
    - *Concept:* Classical ML vs. Deep Learning vs. Generative AI [cite: 21]. Understanding tokenization, embeddings, and context windows [cite: 21].
    - *Hands-on Project Application:* Write a backend routing script to send raw article text data to the OpenAI/Anthropic APIs [cite: 22]. Instruct the LLM to run summarizations, extract primary text claims, and clean up noisy formatting [cite: 22].
- **Week 6: Vector Databases & Semantic Search [cite: 23]**
    - *Concept:* High-dimensional vector spaces and vector databases (Pinecone, Chroma, or pgvector) [cite: 24]. Understanding how AI maps mathematical "geography" to concepts [cite: 24].
    - *Hands-on Project Application:* Programmatically convert the collected text data into high-dimensional vector embeddings and store them within a vector database provider for rapid conceptual search execution [cite: 25].
- **Week 7: Retrieval-Augmented Generation (RAG) Architecture [cite: 26]**
    - *Concept:* Anchoring LLMs to verified, private datasets to eliminate hallucination risks [cite: 27].

    - *Hands-on Project Application:* Build the core RAG inference engine. When a user queries a topic, programmatically find relevant background documents in the vector database, inject those facts directly into an augmented LLM prompt template, and surface structured answers [cite: 28].
- **Week 8: LLM Tool Use & Function Calling [cite: 29]**
    - *Concept:* Teaching AI agents how to systematically interact with external APIs, execute internal code routines, and run dynamic data lookups [cite: 30].
    - *Hands-on Project Application:* Configure an advanced LLM agent that reads a user's natural language request and intelligently determines *when* to execute a SQL query on the relational database vs. when to use semantic search vector lookups [cite: 31].

Phase 3: The Capstone Pipeline & Deployment (Weeks 9–12) [cite: 32]

Goal: Stitch the components into a single production-grade pipeline [cite: 33].

- **Week 9: System Integration & Orchestration [cite: 34]**
    - *Concept:* Connecting the Phase 1 Ingestion engine seamlessly with the Phase 2 AI engine [cite: 35].
    - *Hands-on Project Application:* Automate the fully unified application execution loop. Ensure newly arriving financial and tech news reports are instantly parsed, vectorized, classified for Hype vs. Evidence, and routed directly to the AI agent storage layers [cite: 36].
- **Week 10: Building the Intelligent UI [cite: 37]**
    - *Concept:* Rapid prototyping frameworks built specifically for full-stack data applications [cite: 38].
    - *Hands-on Project Application:* Build an interactive front-end web dashboard using Streamlit or React [cite: 39]. Design search input functionality, trend visualization components, and an integrated conversational chat module to converse with the backend AI system [cite: 39].
- **Week 11: CI/CD & Cloud Deployment [cite: 40]**
    - *Concept:* Continuous Integration/Continuous Deployment pipelines, cloud environment variables management, and Identity & Access Management (IAM) security roles [cite: 41].
    - *Hands-on Project Application:* Package the functional application and deploy it live to a cloud container runtime engine (e.g., AWS ECS or GCP Cloud Run) to expose the live URL to the internet securely [cite: 42].
- **Week 12: Production Monitoring & The Capstone Review [cite: 43]**
    - **EXTRA ATTENTION MODULE: Cost & Latency Optimization** [cite: 44]
        - *Concept:* Budget control, operational observability, application logs, and cost/value mapping [cite: 44]. How much does this AI dashboard cost to operate continually [cite: 44]?
        - *Specialized Learning Items:*
            1. **AI Inference Financial Tracking:** Build a rigorous framework to audit token consumption metrics across API endpoints (Input vs. Output tokens vs. Vector Database operational pricing) [cite: 44].
            2. **Cost Estimate Formulation:** Create a predictive pricing model projecting monthly running costs based on 100, 1,000, and 10,000 daily user queries [cite: 44].

3. **Benefit/Value Analysis:** Quantify engineering trade-offs (e.g., assessing if swapping an expensive, high-performing model for a smaller, fine-tuned open-source model preserves acceptable Hype vs. Evidence score accuracy while lowering costs by 80%) [cite: 44].
4. **Latency Troubleshooting:** Identify bottlenecks in prompt execution chains and configure Redis caching or streaming responses to speed up the dashboard interface.

- *Hands-on Project Application:* Conduct a final comprehensive code base audit, showcase a live end-to-end production deployment demonstration, and compile a technical project retrospective report [cite: 45].

Future Project Extensions (Post-MVP Planning)

Once the baseline MVP architecture operates reliably in production, Isabella can introduce these advanced research concepts:

1. **Market Psychology Indicators:** Deep NLP classification targeting public discussion boards to measure speculative asset bubbles, overconfidence, fear, or rapid-onset hype cycles.
2. **Narrative vs. Evidence Tracking:** Mathematical divergence indexing that explicitly tracks situations where conversational excitement outpaces verifiable factual disclosures.
3. **Contrarian View Discovery:** Automated discovery routing that surfaces alternative, high-evidence dissenting insights when conversational consensus crosses extreme sentiment thresholds.
4. **Narrative Evolution Timelines:** Dynamic timeline graphs logging how underlying theses morph in response to concrete market shocks or quarterly corporate earnings data.